MIL-A-20383D
5 October 1982
SUPERSEDING
MIL-A-20383C
1 April 1977

MILITARY SPECIFICATION

ARROW SETS, SURVEYING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers surveyors' arrow sets, each set consisting of 11 arrows and a carrying ring.

1.2 Classification. Arrows shall be the following types, as specified (see 6.2):

Type I - 10 inches long.
Type II - 14 inches long.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-E-489	- Enamel, Alkyd, Gloss (for Exterior and Interior Surfaces).
PPP-B-636	- Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6675

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MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-S-3534	- Surveying Instruments and Accessories, Packaging of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASTM E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The arrow sets shall each consist of 11 arrows with a metal ring on each arrow to accommodate the arrows on a carrying ring.

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3.2 Material. The material for the arrow sets shall be cold-drawn round steel wire between 3/16 inch and 1/4 inch in diameter, with a minimum Rockwell "B" hardness number of 85 (see 6.3 and 6.4).

3.3 Construction.

3.3.1 Arrows. The top of the arrow shall be formed into a ring in the vertical plane with an inside diameter of 1-1/2 inches, plus or minus 3/8 inch. The lower end of the arrow shall be a pyramidal or conical point, the length of which shall be not less than twice the diameter of the wire. The overall length from the bottom of the ring to the point of the arrow shall be 10 inches, plus or minus 1 inch, for type I arrows, and 14 inches, plus or minus 1 inch, for type II arrows.

3.3.2 Carrier ring. The carrier ring shall be formed into either a 6-inch-diameter loop with the ends bent back in such a manner as to provide a means of hooking the arrows, or a triangular shape with sufficient spring action to permit easy placement and removal of the arrows.

3.4 Treatment and painting. The arrow sets shall be cleaned, treated, and painted as specified in MIL-T-704, type A, The finish coat shall be enamel conforming to TT-E-489, type 1, class B. The color of the arrow ring shall be either gloss red or alternate bands of gloss white and gloss red not less than 1 inch nor more than 2 inches wide. The color of the arrow shank shall be either gloss white or alternate bands of gloss white and gloss red not less than 1 inch wide. The color of the carrier ring shall be gloss white.

3.5 Identification marking. The arrow sets shall be identified in accordance with MIL-STD-130.

3.6 Workmanship. External surfaces shall be free from nonfunctional sharp edges and projecting points.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).

b. Inspection of packaging (see 4.5).

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4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the defects specified in 4.4.1. AQL shall be 2.5 percent defective for major defects, and 4.0 percent defective for minor defects.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2.1. AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The arrow sets shall be examined as specified herein for the following defects:

Major

101. Set not complete.
102. Material not as specified.
103. Construction not as specified.
104. Treatment and painting not as specified.
105. Workmanship not as specified.

Minor

201. Identification marking missing, incomplete, or illegible.

4.4.2 Test.

4.4.2.1 Hardness. The arrow sets shall be tested for hardness in accordance with ANSI/ASTM E 18. Evidence of Rockwell "B" hardness less than 85 shall constitute failure of this test.

4.5 Inspection of packaging. Preservation, packing and marking for level A or B shall be examined and tested in accordance with the Quality Assurance Provisions of MIL-S-3534. Commercial preservation, packing, and marking shall be examined to determine compliance with MIL-STD-1188.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, level B, or commercial, as specified (see 6.2).

5.1.1 Level A. Each arrow set with ring shall be preserved in accordance with the level A preservation and packaging requirements of MIL-S-3534.

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5.1.2 Level B. Each arrow set with ring shall be preserved in accordance with the level B preservation and packaging requirements of MIL-S-3534.

5.1.3 Commercial. Each arrow set with ring shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

5.2.1 Level A. Arrow sets of like description, preserved as specified in 5.1, shall be packed together in accordance with the level A packing requirements of MIL-S-3534.

5.2.2 Level B. Arrow sets of like description, preserved as specified in 5.1, shall be packed together in accordance with the level B packing requirements of MIL-S-3534 except boxes conforming to PPP-B-636, V3c, style optional, may also be used. Box closure and sealing shall be in accordance with the appendix to the box specification.

5.2.3 Commercial. Arrow sets, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The arrow sets are intended for use by surveyors for the temporary marking of taped distances. The arrows are sometimes called chaining pins.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of arrow sets required (see 1.2).

c. Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

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6.4 Materiel deterioration prevention and control (MADPAC). The contracting officer should consider for inclusion in all applicable invitations for bid, requests for proposal, and contracts the following statement:

"The technical proposal shall provide assurance that all system/item parts shall be fabricated from compatible materials, suitably finished, to provide protection from corrosion and other forms of materiel deterioration. The assurance shall be in the form of a finish plan, utilizing state-of-art technology, which clearly identifies the specific finish or treatment to be used on the various components and assemblies, to provide protection against the various forms of materiel deterioration in the applicable operating and storage environments to which they may be exposed."

Custodians:
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Navy - YD
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Review activity:
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